

Dipto Kumar Haldar

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in Dipto Kumar Haldar

🎓 Google Scholar

Education

2025	Khulna University of Engineering and Technology (KUET) Bachelor of Science in Industrial and Production Engineering CGPA: 3.35 / 4.00 Key Undergraduate Courses: Operations Management, Operations Research, Quality Management, Logistics and Supply Chain Management, Project Management
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Research Interests

Supply chain optimization; operations research; multi-hazard risk assessment; mathematical modeling; data-driven decision making; optimization under uncertainty; machine learning; community resilience decision support.

Research Experience

2026-Present	Research Affiliate Risk-Informed Systems Engineering Lab (RISELAB) Ongoing Research Project: Equitable Disaster Resource Allocation under Causal Ambiguity Developed a novel robust optimization framework combining causal inference, partial identification, and predictive analytics to improve disaster resource allocation under uncertainty. The project addressed hidden confounding in evacuation behavior and introduced a fairness-driven allocation strategy under incomplete information. Applied the framework to Hurricane Ian evacuation data across 223 census tracts in Florida, quantifying trade-offs between efficiency, equity, and robustness. The work contributes to resilient emergency planning by advancing counterfactual policy evaluation and identification-aware decision-making. Supervisor: Himadri Sen Gupta, Ph.D. Assistant Professor, Colorado State University Pueblo
2025	Undergraduate Thesis Thesis Title: An Analytical Study on Integrating Artificial Intelligence and Industry 4.0 Technologies for Supply Chain Resilience and Sustainability. - Examine how predictive analytics and Industry 4.0 improve supply chain resilience and sustainability. - Utilize quantitative analysis to examine the relationships between technology adoption and sustainable performance. - Identify supply chain resilience as a key mediator linking technology use and sustainability outcomes. - Contribute to the literature on how digital technologies shape sustainable supply chain transformation. - Provide insights for firms in developing economies to strengthen adaptability and competitiveness. Thesis Supervisor: Palash Saha Khulna University of Engineering and Technology (KUET)

Research and Publications

Published Journal Articles

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| 2026 | <ul style="list-style-type: none">Dipto Kumar Haldar*, Palash Saha, H. M. Belal, Oliver G Kayas. An Analytical Study on Integrating Artificial Intelligence and Industry 4.0 Technologies for Supply Chain Resilience and Sustainability. <i>Supply Chain Analytics</i>, 2026.
https://doi.org/10.1016/j.sca.2026.100209 |
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Journal Manuscripts Under Review

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| 2026 | <ul style="list-style-type: none">Dipto Kumar Haldar*, Himadri Sen Gupta. Evaluating Behavioral-Risk Models for Hurricane Evacuation Allocation. <i>Risk Analysis</i>, 2026. Submission ID: 3373113 |
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2026

- **Dipto Kumar Haldar***, Himadri Sen Gupta,Lava Kumar Polu.Equitable Evacuation Support via Conformal-Robust Optimization. *Transportation Research Part E*, 2026.
Submission ID: TRE-D-26-02168

Skills

- **Programming:** Python (NumPy, Pandas, Scikit-learn), Basic C
- **Optimization Solver:** Gurobi, Pyomo
- **Design Software:** AutoCAD, SolidWorks, Fusion 360
- **Data Analysis Software:** SmartPLS, IBM SPSS, Microsoft Office, Excel, Power BI
- **Soft Skills:** Analytical thinking, problem solving, teamwork, communication, leadership, time management

Linguistic Proficiency

- **English:** Fluent Working Proficiency
- **Bengali:** Native Language

Academic Project

2023 **Undergraduate Project**
Khulna University of Engineering and Technology (KUET)
Project Title: “Developed an advanced fire detection and automated notification system using Arduino, GSM Module, and IR sensors”

Industrial Experience

2023 **Industrial Training – BSRM**
Completed industrial training in the Production Department at BSRM. Learned about steel production processes and professional-level industrial operations.
Key Learnings: Safety and organizational goals, furnace and LRF operation, CCM process, spectro lab inspection, maintenance, and utility management.

Extra-Curricular Activities

2024 **Aspire Institute**
Aspire Leadership Program
Developed skills in communication, leadership, and international networking.

2023-2024 **IEM Debating Club (IDC), KUET**
Vice-President
Organized “Multiverse Open 2024.” Strengthened skills in public speaking, critical thinking, and event management.

Achievements & Awards

- **Finalist:**Walton Laptop Presents MindSparks 2022 (Business Case Competition), AUST Innovation and Design Club
- **Semi-finalist:**Hult Prize at KUET 2024 (On-Campus)
- **Semifinalist:**Face the Case Competition (FTC 3.0), IEEE Student Branch CUET, 2023
- **Technical Scholarship:**Awarded based on academic CGPA at KUET
- **Government Scholarship:(SSC Level)** Awarded by the Board of Intermediate and Secondary Education, Jashore, Bangladesh, for outstanding academic performance